Yuancheng (Remo) Shen

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Education

- **New York University** — New York, United States
  Ph.D. in Computer Science — *Sep. 2024 – present*
  Advisor: Prof. Robert Krueger
- **Shandong University** — Shandong, China
  Master in Computer Science — *Sep. 2021 – Jun. 2024*
  Advisor: Prof. Yunhai Wang
- **Jiangsu University of Science and Technology** — Jiangsu, China
  Bachelor in Computer Science — *Sep. 2017 – Jun. 2021*
  GPA: 3.85; Ranking: 2/102

Research Interests

My research interests lie primarily in ***visualization, visual analytics, and human-computer interaction***. I am developing *human-centered visual analytics systems, frameworks, and libraries* for exploring and interpreting ***large-scale, high-dimensional data***, with a particular focus on *biomedical imaging*. My research also explores ***human-LLM collaboration*** in interactive visualization workflows to support expert-guided exploration, interpretation, and scientific discovery.

Publications

[P1] **Y. Shen**, C. Kalsi, P. Motarwar, D. Annuar, J. Stimes, S. Sterling, S. Qiu, I. Illa-Bochaca, I. Osman, D. Fenyo, G. J. Baker, E. Hernando, K. Ruggles, A. Tsirigos, and R. Krueger. CellGalaxy: Multiscale Exploration of Biomedical Image Embeddings. In ***IEEE Transactions on Visualization and Computer Graphics*** (Under Review).

[P2] Yunge Wen, Yaluo Wang, **Yuancheng Shen**, Robert krueger, Paul Liang. PaintCopilot: Modeling Painting as Autonomous Artistic Continuation. ***ACM CHI 2027*** (Under Review).

[P3] C. Kalsi, H. Fathollahian, L. Reichman, S. Gaupp, **Y. Shen**, A. Shabanova, A. Junquera, A. Szabó, A. Rodríguez, T. Vallius, M. Rogava, G. E. Marai, M. Krone, S. Boorboor, and R. Krueger. BioSET: Interactive Visual Analysis of Biomarker Co-localization in Large Multi-Volume Tissue Data. In ***IEEE Transactions on Visualization and Computer Graphics*** (Under Review).

[P4] Yunge Wen, **Yuancheng Shen**, Robert krueger, Paul Liang. PaintCopilot: Modeling Painting as Autonomous Artistic Continuation. In ***NeurIPS Creative AI, 2026.***

[P5] C. Kalsi, **Y. Shen**, S. Gaupp, L. Reichman, M. Rogava, M. Krone, S. Boorboor, and R. Krueger. BioSET - Biomarker-based Spatial co-Expression analysis in Tumor environments. In ***Bio+MedVis Challenge @ IEEE VIS, 2025*** (**Award of Excellence**).

[P6] **Yuancheng Shen**, Yue Zhao, Yunhai Wang, Tong Ge, Haoyan Shi and Bongshin Lee. Authoring Data-Driven Chart Animations through Direct Manipulation. In ***IEEE Transactions on Visualization and Computer Graphics, 2024*** (DOI: 10.1109/TVCG.2024.3491504).

[P7] Tao Dai, Qi Wang, **Yuancheng Shen**, Shang Gao. SwinVision: detecting small objects in low-light environments. In ***IEEE Access, 2025*** (DOI: 10.1109/ACCESS.2025.3548151).

Research Experience

- **Human-in-the-Loop Exploration of Large-Scale, High-Dimensional Data** — New York University
  Advisor: Prof. Robert Krueger — *Sep. 2024 – Present*
  - **Human-AI Collaborative Exploration of Biomedical Image Representations**: Exploring *large-scale, high-dimensional biomedical image embeddings* poses challenges in scalability, interpretability, and connecting learned representations to their original tissue context. To address these challenges, I developed *CellGalaxy*, a scalable visual analytics system that supports interactive exploration of up to *1.5 million image patches* through *multiscale visualization*, *semantic zooming*, and coordinated views of embedding spaces and spatial tissue organization. We further integrate *LLM-generated semantic explanations* with *human-in-the-loop validation and refinement*, enabling biomedical experts to collaboratively interpret complex cellular patterns and support scientific discovery.
  - **Guided Visual Analytics of Large-Scale 3D Tissue Imaging**: Exploring *terabyte-scale, multi-channel 3D tissue data* poses challenges in identifying spatially localized biomarker relationships. To address these challenges, we developed *BioSET*, combining *multiscale volume visualization*, set-based *biomarker co-localization analysis*, and spatial guidance to help experts identify and inspect regions of interest. We further integrate *VLM-assisted semantic explanations* and interactive bookmarking to support expert-guided interpretation and scientific analysis.
  - **Representation Learning for 3D Cellular Morphology and Spatial Context**: We investigate methods for learning meaningful representations from *3D cellular microscopy data*, where complex morphology and spatial tissue organization are difficult to capture with conventional image features. Our ongoing research explores *vision transformer-based representation learning* to model cellular morphology and spatial context, with the goal of supporting downstream tasks such as *cell registration*, similarity search, and cross-sample comparison.

- **Mixed-Initiative Human-AI Collaboration for Digital Painting** NYU & MIT
  Advisors: Prof. Robert Krueger and Prof. Paul Liang *Apr. 2026 – Sep. 2026*
  - Existing neural painting methods typically rely on *predefined target images*, limiting AI's ability to participate in *open-ended creative processes*. To address this challenge, we developed *PaintCopilot*, a human-AI co-creative painting system that models *autonomous artistic continuation* from an evolving canvas and stroke history. We introduce three lightweight models for predicting evolving artistic intent, generating subsequent brushstrokes via *flow matching*, and sampling region-specific stroke sequences. Through *interruptible and revisable AI assistance*, the system enables artists to dynamically negotiate the scope of AI participation while retaining *creative agency*.

- **Human-Centered Authoring of Data-Driven Chart Animations** Shandong University
  Advisors: Yunhai Wang and Bongshin Lee *Oct. 2022 – Jun. 2024*
  - Creating expressive data-driven chart animations is challenging because existing tools either rely on *predefined templates* or *complex programming and manual keyframe manipulation*. To address these limitations, we developed *CAST+*, an interactive authoring system which combines *keyframe specifications*, *direct manipulation*, and *data-driven auto-completion*. By extending the underlying declarative animation grammar (*Canis*) and introducing *generalized selection-based recommendations*, CAST+ enables users to construct and refine complex keyframe sequences with fewer manual interactions, making expressive chart animation authoring more accessible to non-programmers.

- **Learning-Based Visual Detection in Challenging Environments** Shandong University
  Advisor: Shang Gao *Jan. 2024 – Sep. 2024*
  - **Transformer-based small object detection**: We investigated deep learning approaches for detecting small objects in challenging low-light environments, where weak visual signals and noise significantly degrade object recognition. We propose a transformer-based detection framework that enhances global contextual modeling and multi-scale feature representation, integrating low-light enhancement and attention mechanisms to improve detection accuracy while maintaining computational efficiency.

Selected Honors and Awards

- Best Poster Award, European Society for Spatial Biology (ESSB) *Oct. 2026*
- Award of Excellence, Bio+MedVis Challenge, IEEE VIS *Nov. 2025*
- NYU School of Engineering (SOE) PhD Fellowship, New York University *Sep. 2024*
- Outstanding Graduates and Thesis Award, Shandong University *Jun. 2024*
- Postgraduate Excellent Student Award Fund, Shandong University *Oct. 2021*
- Outstanding Graduates and Thesis Award, Jiangsu University of Science and Technology *Jun. 2021*
- 1st Prize Scholarship, Jiangsu University of Science and Technology *Oct. 2019*
- 1st Prize in Higher Mathematics Competition, Jiangsu *Aug. 2018*

Academic Service

- **Reviewer**: IEEE Vis; ACM CHI; IEEE Pacific Vis; Expert Systems With Applications; PeerJ Computer Science

Academic Engagements

- **European Society for Spatial Biology (ESSB 2026)** Sitges, Spain
  Presented CellGalaxy poster; contributed to BioSET poster (Best Poster Award). *Sep. 2026*

- **IEEE VIS 2025** Vienna, Austria
  Presented a full paper on data-driven chart animation authoring. *Nov. 2025*

- **International Conference on Geometric Modeling and Processing** Qingdao, China
  Organizing team member; assisted with conference coordination and logistics. *Jun. 2024*

- **Geometric Design and Computing Conference** Qingdao, China
  Organizing team member; assisted with conference coordination and logistics. *Aug. 2022*

- **China Visualization and Visual Analytics Conference** Xining, China
  Conference attendee. *Jul. 2022*

Technical Skills

- **Programming**: Python, TypeScript, JavaScript, C++, SQL, R
- **AI & Machine Learning**: PyTorch, Hugging Face Transformers, Vision Transformers, LLM APIs
- **Visualization & Web**: WebGL, Three.js, VTK.js, D3.js, React
- **Scientific Computing**: NumPy, Pandas, SciPy, scikit-learn, OpenCV
- **Development Tools**: Git, Linux, LaTeX, Adobe Illustrator